

Based on your prior experience/ How the interview goes, you will be onboarded either as an Intern / Full time Employee !

AI/ML Enabled Software Dev-Assignment

Assignment 1:

Build an AI-Powered Customer Support Insight Platform

A mid-sized e-commerce company receives **thousands of customer support messages daily** through chat, email, and tickets. The company struggles to understand:

- What customers are complaining about
- Which issues are increasing
- Which problems affect revenue the most
- How support agents should respond

You are asked to build a **small AI-powered system** that ingests support conversations and produces actionable insights for the business.

Dataset

Use any **public dataset of customer support tickets** or generate a synthetic dataset using AI.

Example fields:

ticket_id
timestamp
customer_id
channel (chat/email/web)
message
agent_reply

product
order_value
customer_country
resolution_status

Dataset size: **5k–50k rows**. (Example : <https://www.kaggle.com/datasets/venky73/spam-mails-dataset>)

Requirements

1. AI / Data Understanding

Create a system that:

- Categorizes tickets into **problem categories**
- Detects **sentiment or frustration level**
- Extracts **top recurring issues**
- Generates **suggested responses for agents**

The solution may use:

- LLM APIs
- embeddings
- vector databases
- classical ML
- prompt engineering

Explain **why you chose your approach**.

2. Software Development

Build a **working application** that includes:

Backend

- API for uploading tickets
- API for retrieving insights
- AI processing pipeline

Frontend (lightweight)

A simple dashboard that shows:

- top issues
- sentiment trends
- ticket summaries
- suggested responses

Stack is your choice.

Examples:

- Python + FastAPI
- Node + Express
- Streamlit
- React dashboard

3. Data Pipeline

Implement a basic pipeline:

Raw data → cleaning → enrichment → AI analysis → storage

Requirements:

- structured database
- ability to process **new incoming tickets**
- reproducible pipeline

Bonus:

- batch vs streaming design
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4. DevOps & Deployment

Deploy the system.

Minimum requirements:

- Dockerized application
- cloud deployment (AWS/GCP/Azure/Vercel/Fly/etc)
- CI/CD pipeline

Bonus:

- monitoring
 - logging
 - auto-scaling design
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5. Business Thinking

Produce a short document answering:

1. Which **3 insights** would matter most to leadership?
 2. How could this system **reduce support costs**?
 3. How could it **increase revenue or retention**?
 4. What **metrics** should the company track?
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Deliverables

1. GitHub repository
 2. Running deployed demo
 3. Architecture diagram
 4. 5–10 minute demo video
 5. Short design document explaining:
 - AI choices
 - data model
 - scalability
 - tradeoffs
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Evaluation Criteria

Area	What We Look For
Engineering	code quality, modular design

Data Skills data cleaning, structuring, insights

AI Usage smart use of LLMs/embeddings

DevOps containerization & deployment

Business Thinking ability to connect tech to impact

Bonus Challenges (Optional)

- RAG-based knowledge assistant for support agents
- anomaly detection on spikes in complaints
- multilingual ticket handling
- cost optimization for LLM usage
- automated weekly insight report